ERM
Advanced Telematics



eData Simple

CAN Engine Communication Protocol

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1 General

eData is a dedicated CANBUS interface, able to read/inquire CAN data using various CANBUS protocols. eData is using ERM's CAN Engine technology in order to do that. In eData Simple, CAN Engine comes with predefined and preset protocols and cannot be programmed to additional ones. In order to use additional protocols or customize the eData abilities, the eData Pro firmware needs to be used.

eData Simple supports the following protocols:

- CANBUS: FMS/J1938, CANBUS
- Dynamic OBD: OBD2, VPW, PWM, K-Line

2 Command Structure

All the commands sent to and from the CAN Engine component have an almost uniform header that contains flags for parsing the message.

The structure of the message header is as follows:

Device ID – indicates the type of unit (this number is a default value and can be changed)

Separator – ","

Message type – indicates the type of message

Separator – ","

Data Format – indicates the format of the message data:

A value of 0 indicates the data is in text format (each character is represented by two bytes of type CHAR ASCII).

A value of 1 indicates the data is in binary format (each character is represented by a single byte).

Separator – ","

Is Last Message – indicates whether this message is the last message:

A value of 0 indicates this is not the last message, and additional messages are expected.

A value of 1 indicates this is the last message, and no further messages are expected.

Separator – ","

Message Number – a number indicating the placement of this message within the group of expected messages.

Separator – ","

Configuration ID – contains the configuration number programmed into the unit, and sent in every message transmitted by the unit. **This field does not exist in messages sent to the unit.**

Separator – ","

Message Data

Following is an example of a message sent to the unit:

30,1,19,0,19,,302,0,1,1,03E903E8000009090100010101 0F00FFFFFFFEFFFFFFFEFFFF800008 Following is an example of a message sent from the unit:
19,0,407,0,1,1,1001,03E903E8000009090100010101 0F00FFFFFFFEFFFFFFFEFFFF800008

2.1 Sending & receiving multiple parameters, rules and histograms in a message

When sending or receiving a message to/from the CAN Engine, you can either send/receive a single parameter, rule, or histogram, or you can send/receive multiple parameters, rules and histograms. To indicate this, one byte at the beginning of the message describes the message type, and then we add concatenated same-type structures of parameters, rules, or histograms.

2.2 Updating and deleting parameters, rules and histograms

Update command:

Any time an Update command is sent and the parameter, rule or histogram already exist in the unit, their values are updated with the new values sent by the server, and all other definitions remain unchanged. If the parameter, rule or histogram do not exist in the unit, they are added to the list provided the permitted quantity is not exceeded (if there is no room to store them, the unit returns NCK).

Delete and Update command:

Any time a command is sent to delete and update parameters, rules or histograms, the system first deletes everything that already existed, and then updates with the new values. For example, if a delete and update parameters command is sent, the unit deletes all parameters defined in the unit, and saves only the new parameters it receives.

Command-type ID is included in the binary data of the message:

Every command sent to or from the CAN Engine contains the message type in the first bye of the message's binary data. Commands not intended for the CAN Engine do not contain the message type in the first bye of the message's binary data.

Message Types:

01	Delete and update parameters
02	Update parameters
03	Delete and update rules
04	Update rules
05	Delete and update histograms
06	Update histograms

07	Load all parameters stored in the unit
08	Load all rules stored in the unit
09	Load all histograms stored in the unit
0A	Delete all parameters stored in the unit
0B	Delete all rules stored in the unit
0C	Delete all histograms stored in the unit
0D	Load a specific parameter
0E	Load a specific rule
0F	Load a specific histogram
10	Delete a specific parameter stored in the unit
11	Delete a specific rule stored in the unit
12	Delete a specific histogram stored in the unit
13	Report a rule that occurred in the unit
14	Periodic reporting of parameter values
15	Reporting parameters as a result of a rule that occurred in the unit
16	Reporting parameters after ignition shut off
17	Periodic reporting of histogram values
18	Reporting histograms after ignition shut off
30,1,19,0,19	Error reporting
1A	Erase errors
1B	Erase diagnostic codes from the vehicle
1C	Send VIN of the vehicle
1D	Send parameters after an error code occurs
20	Report after an FMS error in the vehicle
21	Report after an FMI error in the vehicle

2.3 "301" – Sending Commands to the CAN Engine

All commands sent to the CAN Engine should always have 301 as an ID.

The relevant message type (loading parameters, updating parameters, deleting parameters, etc.) will be included in the command data.

2.3.1 Sending Parameter Definitions to the CAN Engine

This section describes the structure of the parameters sent from the server to the CAN Engine for processing.

Example of a command to delete all parameters in the unit and update only two parameters:

30,1,19,0,19,301,0,1,1,0100010000000100080001000 1FFFF000000000020000A 00020000000200 0800020002000200000000120000A

Example of a command for updating a single parameter:

30,1,19,0,19,301,0,1,1,0200010000000100080001000 1FFFF000000000020000 A

Command structure:

Field Name	Size in bytes	Comments
Message Type	1	01 = Delete and update parameters 02 = Update parameters
Parameter ID	2	Defines the number of the current parameter.
Header ID	4	Defines the ID of the message being broadcast for the current parameter.
Start Bit	1	Defines the start bit position of the relevant data within the message.
Bit Count	1	Defines the number of bits used by the relevant data in the message.
Multiplying	2	Defines the number with which to multiply the value extracted from data, for the purpose of calculating the true value.
Divide	2	Defines the number with which to divide the value extracted from the data (after multiplying), for the purpose of calculating the true value.
Offset	2	Defines the number to add to the values extracted from the data (after multiplying and dividing), for the purpose of calculating the true value
Filter Mask	2	Value = Raw-Data & Mask
Filter Value	2	The filter value
Type	1	Defines the type of parameter: **0** – **Regular** type. **1** – **Counter**: Counts the number of times the parameter value changes from zero to one. **2** – **Difference**: The difference between the current value and the previous value (a negative value indicates the parameter decreased, a positive value indicates the parameter increased). **3** – **Absolute Difference**: The absolute difference between the current value and the previous value. **4** – **Accumulated Time**: The accumulated time during which the parameter was active (non-zero) during the trip, in milliseconds. **5** – **Odometer**: For the purpose of updating the StarLink unit with the latest odometer reading received from the vehicle. **6** – **Sum**: The sum of all values received during the trip. **7** – **VSS**: A unique parameter for extracting the VSS value from the vehicle. Will always contain the latest VSS value received. **8- DifferenceSum-** A parameter that contains the sum value of the parameter differences received during the trip. **Speed** - Updates the StarLink unit with the vehicle's last odometer value**.**
Parameter Flags	1	The various flags include: **Bit 7** – Defines whether the parameter is received from an FMS network. **Bit 6** – Defines whether the data is coded in Little-Endian order. **Bit 5** – Defines whether to broadcast this parameter using the Dashboard mechanism. **Bit 4** – Defines whether the parameter value is non-negative or can be negative. **Bit 3** - Defines whether the parameter is reset when the ignition status changes.
Send Parameter Time Interval	2	Defines how often, in seconds, to broadcast this parameter.

2.3.2 Sending Rule Definitions to the CAN Engine

This section describes the structure of rules sent from the server to the CAN Engine for processing.

Example of a command to delete all the rules in the unit and update only two rules:

30,1,19,0,19,301,0,1,1,0300011AAA1BBB00640707021400010E0000022AA A2BBB03E8120D0207021400040F1F4009021400040F3A98

Example of a command for updating one rule in the unit:

30,1,19,0,19,301,0,1,1,0400011AAA1BBB00640707021400010E00

Command structure:

Field Name	Size in Bytes	Notes
Message Type	1	03 = Delete and update rules 04 = Update rules
Rule ID	2	Defines the number of the current rule.
Event ID When Rule Became True	2	Defines the event ID broadcast to the server when the system determines that the rule conditions are occurring.
Event ID When Rule Became false	2	Defines the event ID broadcast to the server when the system determines that the rule conditions stopped occurring.
Debounce Time + Activate Dashboard When Rule Occurs Flag	2	Defines the minimum time (in milliseconds) that the rule must occur or stop occurring, for it to be broadcast to the server. The minimum value is 1. The maximum value is 32767 (32 seconds) Bit no. 15 determines whether to turn on the dashboard when a rule condition occurs.
Rule Data Length	1	The length of the rule, in number of bytes.
Rule Data	Up to 128 bytes	

2.3.3 Sending Histogram Definitions to the CAN Engine

This section describes the structure of histograms sent from the server to the CAN Engine for processing.

Example of a command to delete all the histograms in the unit and update only two histograms:

30,1,19,0,19,301,0,1,1,0500010005FFFFFE0C000001F403000A020013002A0 0020023FFFFEC78 000013880A00140300180021001B

Example of a command for updating one histogram in the unit:

30,1,19,0,19,301,0,1,1,0600010005FFFFFE0C000001F403000A020013002A

Command structure:

Field Name	Size in Bytes	Notes
Message Type	1	05 = Delete and update histograms 06 = Update histograms
Histogram ID	2	Defines the number of the current histogram.
Main Parameter ID	2	Defines the ID of the parameter on which the histogram is based.
Start Histogram Range	4	Defines the beginning of the histogram range.
End Histogram Range	4	Defines the end of the histogram range.
Total Histogram Columns	1	Defines how to divide the histogram range (number of histogram columns).
Send Histogram Time Interval	2	Defines how often, in seconds, the system should broadcast histogram values.
Total Related Parameters	1	Defines the number of parameters related to the current histogram.
Related Parameters ID's List	2 bytes * the number of parameters related to the histogram	Contains the list of IDs of all related parameters.

2.3.4 Loading All Parameters from the CAN Engine

This section describes the structure of the command instructing the CAN Engine to broadcast all the parameters stored in the unit.

Example:

30,1,19,0,19,301,0,1,1,0703E8

Command structure:

Field Name	Size in Bytes	Notes
Message Type	1	07 = Load all parameters stored in the unit.
Max Message Size	2	Defines the maximum message size.

2.3.5 Loading All Rules from the CAN Engine

This section describes the structure of the command instructing the CAN Engine to broadcast all the rules stored in the unit.

Example:

30,1,19,0,19,301,0,1,1,0803E8

Command structure:

Field Name	Size in Bytes	Notes
Message Type	1	08 = Load all rules stored in the unit.

Max Message Size	2	Defines the maximum message size.

2.3.6 Loading All Histograms from the CAN Engine

This section describes the structure of the command instructing the CAN Engine to broadcast all the histograms stored in the unit.

Example:

30,1,19,0,19,301,0,1,1,0903E8

Command structure:

Field Name	Size in Bytes	Notes
Message Type	1	09 = Load all histograms stored in the unit.
Max Message Size	2	Defines the maximum message size.

2.3.7 Deleting All Parameters from the CAN Engine

This section describes the structure of the command instructing the CAN Engine to delete all parameters stored in the unit.

Example:

30,1,19,0,19,301,0,1,1,0A

2.3.8 Deleting All Rules from the CAN Engine

This section describes the structure of the command instructing the CAN Engine to delete all rules stored in the unit.

Example:

30,1,19,0,19,301,0,1,1,0B

2.3.9 Deleting All Histograms from the CAN Engine

This section describes the structure of the command instructing the CAN Engine to delete all histograms stored in the unit.

Example:

30,1,19,0,19,301,0,1,1,0C

2.3.10 Loading a Specific Parameter from the CAN Engine

This section describes the structure of the command used to read a single parameter stored in the unit.

Example:

30,1,19,0,19,301,0,1,1,0D0001

Command structure:

Field Name	Size in Bytes	Notes
Message Type	1	0D = Load a specific parameter stored in the unit.

Parameter ID	2	Defines the ID of the parameter to load.

2.3.11 Loading a Specific Rule from the CAN Engine

This section describes the structure of the command used to read a single rule stored in the unit.

Example:

30,1,19,0,19,301,0,1,1,0E0001

Command structure:

Field Name	Size in Bytes	Notes
Message Type	1	**0E** = Load a specific rule stored in the unit.
Rule ID	2	Defines the ID of the rule to load.

2.3.12 Loading a Specific Histogram from the CAN Engine

This section describes the structure of the command used to read a single histogram stored in the unit.

Example:

30,1,19,0,19,301,0,1,1,0F0001

Command structure:

Field Name	Size in Bytes	Notes
Message Type	1	**0F** = Load a specific histogram stored in the unit.
Histogram ID	2	Defines the ID of the histogram to load.

2.3.13 Deleting a Specific Parameter from the CAN Engine

This section describes the structure of the command used to delete a single parameter stored in the unit.

Example:

30,1,19,0,19,301,0,1,1,100001

Command structure:

Field Name	Size in Bytes	Notes
Message Type	1	**10** = Delete a specific parameter.
Parameter ID	2	Defines the ID of the parameter to delete.

2.3.14 Deleting a Specific Rule from the CAN Engine

This section describes the structure of the command used to delete a single rule stored in the unit.

Example:

30,1,19,0,19,301,0,1,1,110001

Command structure:

Field Name	Size in Bytes	Notes
Message Type	1	11 = Delete a specific rule.
Rule ID	2	Defines the ID of the rule to delete.

2.3.15 Deleting a Specific Histogram from the CAN Engine

This section describes the structure of the command used to delete a single histogram stored in the unit.

Example:

30,1,19,0,19,301,0,1,1,120001

Command structure:

Field Name	Size in Bytes	Notes
Message Type	1	12 = Delete a specific histogram.
Histogram ID	2	Defines the ID of the histogram to delete.

2.3.16 Reading errors from the fault logger CAN Engine

Read the fault codes which are stored at the logger.

Example:

30,1,19,0,19,301,0,1,1, 19

2.3.17 Deleting errors from the fault logger CAN Engine

Delete all the fault codes stored in the logger.

Example:

30,1,19,0,19,301,0,1,1,1A

2.3.18 Deleting fault codes

Delete the fault codes stored in the vehicle ECU (sending a unique command to the vehicle).

Example:

30,1,19,0,19,301,0,1,1,1B

2.3.19 Reading VIN number from the vehicle

Read the value of the VIN number stored in the vehicle.

Example:

30,1,19,0,19,301,0,1,1,1C

2.3.20 Set Car-Type

Set the value of the car type into eData.

Example:

30,1,19,0,301,0,1,1,1E0002

None: For the updated vehicle list please contact our support

2.3.21 Get Car-Type

Read the value of the car type configured in eData.

Example:

30,1,19,0,301,0,1,1,1F

2.3.22 Get Tachograph Driver-ID number from the eData

Read the value of the driver id's stored in the eData.

Example:

30,1,19,0,301,0,1,1,23

2.4 "302" – Transmitting Global Parameters

This section describes the command used to set the Global settings of the unit.

Example:

30,1,19,0,19,**302**,0,1,1,03E903E8000009090101000000010F00130118000303011122
33445566778899A A

Command structure:

Field Name	Size in Bytes	Notes
Configuration ID	2	Specifies the ID of the configuration stored in the unit.
Message Max Size	2	Specifies the maximum size of the message in terms of the number of bytes.
Debounce Time for Ignition On	1	Specifies the stabilization time in seconds of the ignition state when the ignition is turned on.
Debounce Time for Ignition Off	1	Specifies the stabilization time in seconds of the ignition state when the ignition is turned off.
CANBUS # 1 Baud Rate	1	Specifies the Baud Rate of CANBUS # 1: • 0 = 10K • 1 = 20K • 2 = 33.333K • 3 = 50K • 4 = 62.5K • 5 = 8 3.333K • 6 = 100K • 7 = 125K • 8 = 250K 9 = 500K
Process CANBUS while	1	• 1 = Check and process CAN + RULES messages when the ignition is off. 0 = No processing when the ignition is off.

Ignition state is Off		
Send Parameters at Trip End	1	1. Determines whether to send all parameters at the end of the trip.
Send Histograms at Trip End	1	2. Determines whether to send all histograms at the end of the trip.
Send Data Binary	1	Determines the transmission format of the data: • 0 – Text 3. 1 – Binary
CANBUS # 1 In Silent Mode	1	Determines whether CANBUS # 1 works in silent mode or in Send Acknowledge working mode: • 0 – CANBUS sends acknowledgements 1 – CANBUS does not send acknowledgements
CANBUS # 2 In Silent Mode	1	Specifies whether CANBUS # 2 works in silent working mode or in Send Acknowledge working mode: • 0 – CANBUS sends acknowledgements 1 – CANBUS does not send acknowledgements
OBD Unit	1	Determines whether the unit processes the OBD protocol: • 0 – Do not process the OBD protocol 1 – Process the OBD protocol
OBD Protocols Bit Mask	1	Defines which OBD-type protocols the unit should try to process: • Bit 1 – Process OBD CANBUS • Bit 2 – Process OBD VPW • Bit 3 – Process OBD PWM Bit 4 – Process OBD KLINE
Regular Protocols Bit Mask	1	Defines which regular protocols the unit should try to process: • Bit 1 – Process Regular CANBUS Bit 2 – Process J1708
Device ID	1	The ID of the product. (You can connect several units on the sale line) • Minimum value – 1 Maximum value – 255
Report Parameters When Rule Occurs	1	Determines whether the system broadcasts all CAN Engine parameters before broadcasting the rule itself. • 0 – Does not broadcast parameters before the rule 1 – Does broadcast parameters before the rule
Process FMS DTCs	1	If active, the system queries faults from the FMS.
Send Odometer Options	1	Determines the reporting type for the Odometer variable: • 1 km • 100 m • 10 m • 1m • 1 km per trip, • 100 m per trip, • 10 m per trip, 1m per trip
Debounce Count to Accept DTC	1	Determines the minimum number of times a vehicle must receive a fault before reporting a new fault. The value must be ≥ 1.

Debounce Count to Decline DTC	1	Determines the minimum number of times a problem should stop in a vehicle before a fault has been reported from the vehicle (a value must be greater than or equal to 2)
Report Parameters When Fault Code Occurs	1	Determines whether the system transmits all CAN Engine parameters before the fault code is transmitted. • 0 – Not active 1 – Active
Fault Codes to Ignore Array	2*5	A list of fault codes which the system should ignore. There is a maximum of five fault codes to ignore. 2 bytes are used for each fault code. Once a fault code appears in the above list, the system ignores it.

2.5 "303"-Loading All Global Settings Definitions

This section describes the structure of the command used to upload from the unit the definitions of all the Global Settings.

Example:

30,1,19,0,19,**303**,0,1,1,

2.6 "304"-Resetting Global Settings to their Default Values

This section describes the structure of the command used to set all Global Settings to their default values.

Example:

30,1,19,0,19,**304**,0,1,1,

"305"-Resetting the Unit

This section describes the structure of the command used to reset the unit.

Example:

30,1,19,0,19,**305**,0,1,1,

2.7 "306"-Retrieving Diagnostic Data from the Unit

This section describes the structure of the command used to query the unit regarding the types of communications identified by the unit – including the identification of errors on the CAN bus.

Example:

30,1,19,0,19,**306**,0,1,1,

2.8 "307" – Activating the Dashboard Mechanism (High Speed Parameter Reporting)

This section describes the command used to instruct the unit to start broadcasting dashboard-type parameters at the time intervals defined in the command.

Example:

30,1,19,0,19,**307**,0,1,1,012C000A

Command structure:

Field Name	Size in Bytes	Notes
Total Dashboard Time	2	Specifies the total time duration, in seconds, when the Dashboard mechanism will be active.
Time Interval	2	Specifies the time intervals between dashboard messages broadcasts.

2.9 "308" – Loading Version Information from the Unit

This section describes the command used to query the unit regarding its software version and hardware version.

Example:

30,1,19,0,19,**308**,0,1,1,

2.10 "309" – Updating the Unit's Version

This section describes the command used to operate the version upgrade mechanism within the unit.

Example:

30,1,19,0,19,**309**,0,1,1,

2.11 "310" – Enables the transmission of CAN messages on a communication line

When the unit receives this command, it starts broadcasting the command information sent to the vehicle's communication line (CAN).

Example:

30,1,19,0,19,**310**,0,1,1,0100001234081020304050607080000A000A50

Field Name	Example value	Size in Bytes	Notes
CAN Type	01	1	Specifies the type of message HEADER ID. • 0 – Standard 11 Bit. • 1 – Extended 29 Bit
Header ID	00001234	4	Sets the HEADER ID to be broadcast on the line
Message Data Length	08	1	The length of the message (1-8)
Message Data Byte # 1	10	1	The value of Byte number 1 in the message.
Message Data Byte # 2	20	1	The value of Byte number 2 in the message.
Message Data Byte # 3	30	1	The value of Byte number 3 in the message.

Message Data Byte # 4	40	1	The value of Byte number 4 in the message.
Message Data Byte # 5	50	1	The value of Byte number 5 in the message.
Message Data Byte # 6	60	1	The value of Byte number 6 in the message.
Message Data Byte # 7	70	1	The value of Byte number 7 in the message.
Message Data Byte # 8	80	1	The value of Byte number 8 in the message.
Total Retries	000A	2	The number of times the unit transmits the message.
Time Interval	000A	2	The time interval between transmissions in milliseconds.
Message Validation XOR	50	1	The XOR value of all fields (bytes) of the message (for validation purposes) The initial value from which the XOR must be sum is 0xFF

3 Receiving Commands from the Unit

3.1 "401" – Loading CAN Engine Parameters from the Unit

This section describes the format of the data arriving from the unit in response to a request for loading a specific parameter or several parameters from the unit.

Example of one parameter f received from the unit:

19,0,**401**,0,1,1,1001,07000180BD00010008000100010000010001000F0003C

Example of two parameters received from the unit:

19,0,401,0,1,1,1001,07000180BD00010008000100010000010001000F0003C0
002880004E800100 002000AFFCE1001100101A00078

Data structure:

Field Name	Size in Bytes	Notes
Message Type	1	07 – Loading parameters stored in the unit.
Parameter ID	2	
Header ID	4	
Start Bit	1	
Bit Count	1	
Multiplying	2	
Divide	2	

Offset	2
Filter Mask	2
Filter Value	2
Type	1
Parameter Flags	1
Send Parameter Time Interval	2

3.2 "402" – Loading CAN Engine Rules from the Unit

This section describes the format of the data arriving from the unit in response to a request for loading a specific rule or several rules from the unit.

Example of one rule received from the unit:

19,0,402,0,1,1,1001,0800011AAA1BBB00640707021400010E00

Example of two rules received from the unit:

19,0,**402**,0,1,1,1001,0800011AAA1BBB00640707021400010E0000022AAA2BBB 03E8120D020702 1400040F1F4009021400040F3A98

Data structure:

Field Name	Size in Bytes	Notes
Message Type	1	08 – Loading rules stored in the unit.
Rule ID	2	
Event ID When Rule Became True	2	
Event ID When Rule Became false	2	
Debounce Time	2	
Rule Data Length	1	
Rule Data	Up to 128 bytes	

3.3 "403" – Loading CAN Engine Histograms from the Unit

This section describes the format of the data arriving from the unit in response to a request for loading a specific histogram or several histograms from the unit.

Example of one histogram received from the unit:

19,0,403,0,1,1,1001,0900010001FFFFFFFF00000001030001010001

Example of two histograms received from the unit:

19,0,**403**,0,1,1,1001,0900010001FFFFFFFF00000001030001010001000200 02FFFFFFFE0000000 20300010200020003

Data structure:

Field Name	Size in Bytes	Notes
Message Type	1	09 – Loading histograms stored in the unit.
Histogram ID	2	
Main Parameter ID	2	
Start Histogram Range	4	
End Histogram Range	4	
Total Histogram Columns	1	
Send Histogram Time Interval	2	
Total Related Parameters	1	
Related Parameters ID's List	2 bytes * the number of parameters related to the histogram	

3.4 "404"- Rule Reporting Structure

This section describes the format of the data arriving from the unit as a result of a certain rule occurring in the unit.

Example of one occurring rule received from the unit:

19,0,**404**,0,1,1,1001,1300011AAA000000000001

Example of two occurring rules received from the unit:
19,0,404,0,1,1,1001,1300011AAA00000000000100021BBB000000000001

Data structure:

Field Name	Size in Bytes	Notes
Message Type	1	13 – Reporting a rule that occurred in the unit.
Rule ID	2	The ID of the rule that occurred in the unit.
Event ID	2	The ID of the event that occurred in the unit.
Rule Time	4	Defines the total duration, in milliseconds, of the event.
Total Count	2	Defines the number of times the event occurred.

3.5 "405" – Periodic Parameter Reporting Structure

This section describes the data format of the values of parameters defined in the unit to be reported periodically.

Example of reporting the value of a single parameter:

19,0,**405**,0,1,1,1001,140001000155

Example of reporting the value of two parameters:

19,0,**405**,0,1,1,1001,14000100015500020102 1234

Data structure:

Field Name	Size in Bytes	Notes
Message Type	1	**14** – Periodic reporting of parameter values.
Parameter ID	2	The ID of the reported parameter.
Parameter Type	1	The type of reported parameter.
Parameter Value Size	1	Defines the length, in bytes, of the value of the reported parameter (1, 2 or 4 bytes).
Parameter Value	According to the value in Parameter Value Size	The most recent reading of the parameter value in the unit.

3.6 "406" –Parameter Value Reporting Structure as Result of Rule Occurring

This section describes the data format of values of parameters defined in the unit and reported as the result of a rule occurring.

Note that first the parameters are reported and then the rule is reported.

Example of reporting the value of a single parameter:

19,0,**406**,0,1,1,1001,150001000155

Example of reporting the values of two parameters:

19,0,**406**,0,1,1,1001,1500010001550002010212 34

Data structure:

Field Name	Size in Bytes	Notes
Message Type	1	15 – Reporting parameter values as the result of a rule occurring in the unit.
Parameter ID	2	The ID of the reported parameter.
Parameter Type	1	The type of reported parameter.
Parameter Value Size	1	Defines the length, in bytes, of the value of the reported parameter (1, 2 or 4 bytes).
Parameter Value	According to the value in Parameter Value Size	The most recent reading of the parameter value in the unit.

3.7 "407"– Parameter Value Reporting Structure, After Ignition Turned Off

This section describes the data format of values of parameters defined in the unit and reported after ignition turned off (at end of trip).

Example of reporting the value of a single parameter:

19,0,**407**,0,1,1,1001,160001000155

Example of reporting the values of two parameters:

19,0,**407**,0,1,1,1001,160001000155000201021234

Data structure:

Field Name	Size in Bytes	Notes
Message Type	1	16 – Reporting parameter values after ignition turned off.
Parameter ID	2	The ID of the reported parameter.
Parameter Type	1	The type of reported parameter.
Parameter Value Size	1	Defines the length, in bytes, of the value of the reported parameter (1, 2 or 4 bytes).
Parameter Value	According to the value in Parameter Value Size	The most recent reading of the parameter value in the unit.

3.8 "408"– Periodic Reporting Structure of Histogram Values

This section describes the data format of values of histograms defined in the unit and reported periodically.

Example of reporting the value of a single histogram consisting of three columns and two related parameters:

19,0,408,0,1,1,1001,17000103020002000300000185FFFFFCF6FFFFFB71000001
8500000000000 000000000018500000030A0000048F

Example of reporting the values of two histograms, each of which consists of three columns and two related parameters:

19,0,**408**,0,1,1,1001,17000103020002000300000920FFFFEDC0FFFFE4A000000
9200000000000000000000009200000124000001B60000203020005000600000
092000002DA0000036C000000920F
FFFD260FFFFC940000009200000000000000000

Data structure:

Field Name	Size in Bytes	Notes
Message Type	1	17 – Periodic reporting of histogram values.
Histogram ID	2	The ID of the reported histogram.
Total Columns	1	The number of columns in the reported histogram.
Total Related Parameters	1	The number of parameters related to the histogram.
Related Parameters Id's List	2 bytes per each parameter ID * the number of parameters related to the histogram	The list of parameter IDs for the parameters related to the histogram.

Column Total Count	4	The number of times the value of the histogram's main parameter was within the range of the column. Note: This field is sent for each column defined in the histogram.
Total Related Parameters * Parameter Value	4 bytes for each parameter * the number of parameters related to the histogram	For each parameter defined for the histogram – the sum of parameter values in this column. If no parameters are defined, this field is not sent. Note: This field is sent for each column defined in the histogram.

Both fields marked with bold are repeated according to the number of columns defined in the histogram.

3.9 "409"– Periodic Reporting Structure of Histogram Values after IGN Switched off

This section describes the data format for histogram values defined in the unit and reported periodically.

Example #1: Reporting the value of a single histogram consisting of three columns and two related parameters:

19,0,409,0,1,1,1001,18000103020002000300000185FFFFFCF6FFFFFB71000001
8500000000000 000000000018500000030A0000048F

Example #2: Reporting the values of two histograms, each of which consists of three columns and two related parameters:

19,0,409,0,1,1,1001,18000103020002000300000920FFFFEDC0FFFFE4A000000
9200000000000000000000009200000124000001B60000203020005000600000
092000002DA0000036C000000920F
FFFD260FFFFC940000009200000000000000000

Data structure:

Field Name	Size in Bytes	Notes
Message Type	1	18 – Periodic reporting of histogram values after the IGN is switched off.
Histogram ID	2	The ID of the reported histogram.
Total Columns	1	The number of columns in the reported histogram.
Total Related Parameters	1	The number of parameters related to the histogram.
Related Parameters ID List	2 bytes per each parameter ID * the number of parameters related to the histogram	The list of parameter IDs for the parameters related to the histogram.
Column Total Count	4	The number of times the value of the histogram's main parameter was within the range of a particular histogram column.

Total Related Parameters * Parameter Value	4 bytes for each parameter * the number of parameters related to the histogram	The sum of all parameter values defined for the histogram. If no parameters are defined, this field is not sent.

Both fields marked with bold are repeated according to the number of columns defined in the histogram.

3.10 "410" – Detailed structure of Global Parameters reports

This section describes the settings and structure arriving from the unit for each global parameter.

Example of an entire row as shown on the server:

19,0,**410**,0,1,1,1001,**03E903E8000009090101000000010F00130118000303011122334455667788 99AA**

Data structure:

Data (HEX)	Field Name	Size in Bytes (byte = 2 characters/digits)
03E9	Configuration ID	2
03E8	Message Max Size	2
00	Debounce Time For Ignition On	1
00	Debounce Time For Ignition Off	1
09	CANBUS # Baud Rate	1
09	CANBUS # 2 Baud Rate	1
01	Send Parameters In Trip End	1
01	Send Histograms In Trip End	1
00	Send Data Binary	1
00	CANBUS # 1 In Silent Mode	1
00	CANBUS # 2 In Silent Mode	1
01	OBD Unit	1
0F	OBD Protocols Bit Mask	1
00	Regular Protocols Bit Mask	1
13	Device ID	1
01	Report Parameters When Rule Occurs	1
18	Keep Alive Message Time Interval	1
00	Send Odometer Options	1

03	Debounce Count To Accept DTC	1
03	Debounce Count To Decline DTC	1
01	Report Parameters When Fault Code Occurs	1
1122334455667788 99AA	Fault Codes To Ignore Array	2*5

3.11 "411" – Detailed structure of Diagnostic message reports

This section describes the settings and structure arriving from the unit for Diagnostic message.

Example of an entire row as shown on the server:

19,0,**411**,0,1,1,1001,**05**01*01*01

Data structure:

Data (HEX)	Field Name	Size in Bytes (byte = 2 characters/digits)	Notes
05	Last Communication Bit Mask	1	Bit 0 → CANBUS Communication
			Bit 1 → j1708 Communication
			Bit 2 → OBD Communication
01	Last OBD Communication Bit Mask	1	Bit 0 → OBD CANBUS 1
			Bit 1 → OBD Kline
01	Have CANBUS Errors	1	1 = Error Online
			0 = No Error Online
01	Ignition State	1	Ignition Is Off = 0
			Ignition Is On = 1
			Ignition State Is Unknown = 2

3.12 "412" – Detailed structure of device Hardware

This section describes the settings and structure arriving from the unit for device Hardware.

Example of an entire row as shown on the server:

19,0,**412**,0,1,1,1001,**04**17*01*02**09**

Data structure:

Data (HEX)	Field Name	Size in Bytes (byte = 2 characters/digits)	Notes

04	Software Major Version	1	
17	Software Minor Version	1	
01	Hardware Major Version	1	
02	Hardware Minor Version	1	

09 (01=Hardware in BitMask + 08=Simple eData)	Hardware Type	1	Idata1E2 Hardware = 1
			CANG1E3 Hardware = 2
			Simple_eData Hardware = 3
			
			u8 Get_Device_Details_BitMask(void)
			{
			u8 Result_BitMask = 0;
			#ifdef IDATA1E2
			Result_BitMask |= (1 << 0);
			#endif
			#ifdef CANG1E3
			Result_BitMask |= (1 << 1);
			#endif
			
			#ifdef PROFFESIONAL_IDATA
			Result_BitMask |= (1 << 2);
			#endif
			#ifdef SIMPLE_IDATA
			Result_BitMask |= (1 << 3);
			#endif
			return Result_BitMask;
			}

3.13 "413" – Detailed Structure of OBD Fault messages

Detail data format of the message containing the faults received from the vehicle (OBD module).

Example of 3 faults as shown on the server:

19,0,**413**,0,1,1,1001,191111016222028333 01

Fault no.1 the code received 1111, original fault code P1111

Fault no.1 the code received 6222, original fault code C2222

Fault no.1 the code received B333, original fault code B3333

Data structure:

Field Name	Size in Bytes (byte = 2 characters/digits)	Notes
Message Type	1	19 – Error report
Fault Code Real ID	2	ID of the reported fault
Fault Code Status	1	Status of the reported fault 01 – The fault is added to the logger 02= the fault was deleted from the logger

3.14 "414" – Detailed structure of VIN Messages

Detail the data format of the message containing the VIN number (Vehicle number)

Example:

19,0,**414**,0,1,1,1001,1C30313233343536373839414243444546 47

Data structure:

Field Name	Size in Bytes (byte = 2 characters/digits)	Notes
Message Type	1	C1= Report of the VIN
VIN Number	17	VIN

3.15 "415" – Detailed Structure of Parameter Values after the Fault was Received

Structure of reporting parameter values after a fault received in the vehicle.

Example of 1 parameter sent to the server:

19,0,**415**,0,1,1,1001,1D0001000155

Example of 2 parameters sent to the server:

19,0,**415**,0,1,1,1001,1D000100015500020102 1234

Data structure:

Field Name	Size in Bytes (byte = 2 characters/digits)	Notes
Message Type	1	**1D** – FMS error report
Parameter ID	2	The reported parameter ID
Parameter Type	1	The type of reported parameter
Parameter Value Size	1	Sets the reported bit length (1,2,4)
Parameter Value	1	The last value the unit received

3.16 "416" – Detailed structure of Car-Type Messages

Detail the data format of the message containing the Car-Type

Example:

19,0,**416**,0,1,1,1001,1F0002

Data structure:

Field Name	Size in Bytes (byte = 2 characters/digits)	Notes
Message Type	1	
Report of the Car-Type	1F	
Car-Type	2	0002

3.17 "417" – Detailed Structure of CAN Disconnection

This event is reported from the unit when the ignition switch is active (IGN ON) and no messages are received for the CAN for 500 milliseconds.

Example of a fault on the server:

19,0,**417**,0,1,1,1001,

3.18 "418" – Detailed Structure of FMS Faults

Detail the data format of the message containing the faults received from the vehicle (FMS module)

Example of 3 faults as shown on the server:

19,0,**418**,0,1,1,1001,2080495256560185484948480 2

Fault no. P1482 was added

Fault no. U0100 was deleted

Data structure:

Field Name	Size in Bytes (byte = 2 characters/digits)	Notes
Message Type	1	20 = FMS error report
Fault Code ID	5	5 ASCII characters for the received fault
Fault Code Status	1	The status of the reported fault: • 01 – The fault was added to the logger • 02 – The fault was deleted from the logger

3.19 "419" – Detailed Structure of FMI Fault Messages

Detail the data format of the message containing the faults received from the vehicle (FMI)

Example of 3 faults on the server:

19,0,**419**,0,1,1,1001,210000A5A605010000B123040 2

Fault no.1 the value SPN = 0000A5A6, and the value FMI = 05 were added

Fault no.2 the value SPN = 0000B123, and FMI = 04 were deleted

Data structure:

Field Name	Size in Bytes (byte = 2 characters/digits)	Notes
Message Type	**1**	**21 –** FMS error report
Fault Code SPN	**1**	The SPN value of the fault (19 bit) The maximum value is 524,287.
Fault Code FMI	**1**	The FMI value of the fault (5 bit) The maximum value is 31.
Fault Code Status	**1**	The status of the reported fault: • 01 – The fault was added to the logger • 02 – The fault was deleted from the logger

3.20 ”420” – Detailed Structure of OBD Fault Messages

The unit reports this event when the ignition switch is off (IGN OFF) and a message is received on the CAN.

Example of a fault as shown on the server:

19,0,**420**,0,1,1,1001,

3.21 ”421” – Detailed Structure of an Event when the CAN Goes Back to Sleep

The unit reports this event when the CAN is woken up when the IGN is switched off and the CAN goes back to sleep.

Example of a fault on the server:

19,0,**421**,0,1,1,1001

3.22 “422” – Detailed structure of HDT Faults

Detail the data format of the message containing the faults received from HDT vehicle (Mercedes truck)

Example:

19,0,422,0,1,1,1001,22000601

Data structure:

Field Name	Size in Bytes (byte = 2 characters/digits)	Notes
Message Type	1	22 = Report of the HDT Fault
Fault Code ID	2	

Fault Code Status	1	The status of the reported fault: • 01 – The fault was added to the logger • 02 – The fault was deleted from the logger

3.23 "423" – Detailed structure of Driver-ID's Messages

Detail the data format of the message containing the 2 Driver-ID's

Example:

19,0,423,0,1,1,1001,2330313233343536373839414243444546

Data structure:

Field Name	Size in Bytes (byte = 2 characters/digits)	Notes
Message Type	**1**	**23** = Report of the Driver-ID
Driver-ID	**16**	support 2 driver-id's , 8 bytes each ID 1st ID 3031323334353637 2nd ID 3839414243444546 When no driver ID exist, eData send zeros

3.24 Scan FMS Parameters

Send command to eData to start FMS auto search parameters. (active in FMS version only)

Example:

30,1,19,0,312,0,1,1

"424" – Detailed structure of Scan FMS Finished Messages

Detail the data format of the message containing the number of parameters found during FMS scanning

Example:

19,0,424,0,1,1,1001,240055

Data structure:

Field Name	Size in Bytes (byte = 2 characters/digits)	Notes
Message Type	**1**	**24** = Report of the Scan finished result
Parameters Found	**2**	

4 ERM Contact Information

ERM Support:

Skype: erm.support

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ERM's YouTube Channel: http://www.youtube.com/ermtelematics

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